



Qualification Pack



Drinking Water Purifier Technician

QP Code: NIO/PSC/Q0101

Version: 1.0

NSQF Level: 3.5

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NIO/PSC/Q0101: Drinking Water Purifier Technician

Brief Job Description

The individual should have the knowledge about water purifier installation and repair process. This course will train in different areas of water purifier repair and maintenance. This course comprises of modules on different parts of water purifier system, water quality testing devices, repairing the common faults, replacing faulty components, dealing with customer etc. and it has wide potential to provide employment to both urban and rural youth of the country.

Personal Attributes

The job requires the individual to possess strong technical aptitude, good mechanical skill, proficiency in using hand and power tools along with the ability to work with various plumbing and electrical equipments. They should be dedicated, responsible, trustworthy, attentive, honest and ethical.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIO/PSC/N0101: Install water purifier system \(Domestic and Commercial\)](#)
2. [NIO/PSC/N0102: Perform service and maintenance of Water purifier system \(Domestic and Commercial\)](#)
3. [NIO/PSC/N0103: Effective Resource Utilization](#)
4. [NIO/PSC/N0104: Maintain health and safety at work place](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Plumbing
Sub-Sector	Engineering and Technology
Occupation	Water purifier technician
Country	India
NSQF Level	3.5
Credits	14



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7421.0701
Minimum Educational Qualification & Experience	11th Class (pass) OR Completed 1st year of 3-year diploma (after 10th) and pursuing regular diploma OR 10th grade pass (and pursuing continuous schooling) OR 8th grade pass (with two years of NTC and one year of experience required) OR 8th grade pass with 1 year NTC plus 1 year NAC with 1 Year of experience OR 8th grade pass with 1 year NTC plus 1 year CITS with 1 Year of experience OR 8th grade pass with 3 Years of experience OR Previous relevant Qualification of NSQF Level (Assistant Plumber General- Level-3) with 1-2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	23/06/2028
NSQC Approval Date	23/06/2023
Version	1.0
Reference code on NQR	QG-3.5-PU-00566-2023-V1-NIOS
NQR Version	1



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NIO/PSC/N0101: Install water purifier system (Domestic and Commercial)

Description

This unit talks about Installation of water purifier system in domestic and commercial setup.

Scope

The scope covers the following :

- This unit/task covers the following:
- Pre-installation activities
- Site Preparation,
- Electrical Connections,
- Installing Pre-Filters and Post-Filters

Elements and Performance Criteria

Pre-installation activities

To be competent, the user/individual on the job must be able to:

- PC1.** Evaluate the quality of the water source intended for purification
- PC2.** select the most suitable water purifier system for specific needs
- PC3.** Select the appropriate purification technology and system capacity
- PC4.** Consider factors such as the type of contaminants to be removed, the desired level of purification, water consumption rate, and available space for installation. Properly sizing the system ensures optimal performance and efficiency.
- PC5.** Evaluate the available options and select the most suitable water purifier for your needs..
- PC6.** Determine if any permits or approvals are required for installing a water purifier system in your area
- PC7.** Follow regulations or codes that need to be followed
- PC8.** Contact the appropriate local authorities or agencies to understand the legal requirements and ensure compliance if any
- PC9.** Identify the best location for installing the water purifier system.
- PC10.** Explain factors such as proximity to the water source, access to plumbing connections, availability of electrical outlets (if required), drainage options, and convenient access for maintenance and filter replacements
- PC11.** Ensure that there is adequate space around the installation location for easy access during installation and future maintenance activities
- PC12.** Allow for sufficient clearance to connect plumbing lines, electrical connections, and access the system for servicing and heat dissipation.
- PC13.** Ensure that these utilities are functioning properly and meet the requirements of the water purifier system being installed.
- PC14.** Understand the installation instructions and guidelines provided by the manufacturer of the water purifier system. These instructions may include specific requirements, recommendations, or precautions that need to be followed during installation



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- PC15.** Ensure that all required components and materials for installation, such as plumbing fittings, connectors, pipes, electrical wiring, and mounting hardware, are procured and readily available before starting the installation process.

Site Preparation

To be competent, the user/individual on the job must be able to:

- PC16.** Prepare the installation site by ensuring it is clean, clear of any obstructions, and suitable for accommodating the water purifier system.
- PC17.** Remove debris, organize the area, and provide sufficient space for the system and associated components
- PC18.** Locate and shut off the main water supply
- PC19.** Connect the water purifier system to the existing plumbing lines.
- PC20.** Mount the water purifier unit securely

Electrical Connections

To be competent, the user/individual on the job must be able to:

- PC21.** Ensure proper wiring is in place.
- PC22.** Connect power cables, and install switches or outlets as per the manufacturer's guidelines.

Installing Pre-Filters and Post-Filters

To be competent, the user/individual on the job must be able to:

- PC23.** Install any pre-filters or post-filters that are part of the water purifier system.
- PC24.** Follow the manufacturer's instructions for their installation.
- PC25.** Set up the necessary drainage system for the water purifier unit, especially if it utilizes technologies such as reverse osmosis (RO).
- PC26.** Conduct testing the system to ensure its proper functionality.
- PC27.** Perform calibration of any necessary settings, such as flow rate or pressure regulators, is performed at this stage.
- PC28.** Provide the user with instructions on operating and maintaining the water purifier system.
- PC29.** Explain how to use the control panel (if applicable), replace filters, perform routine maintenance, and troubleshoot common issues.
- PC30.** Demonstrate proper usage to ensure the user's understanding.
- PC31.** After the installation is finalized, clean up the installation area, removing any debris, packaging materials, or tools used during the process.
- PC32.** Restore any disrupted areas, such as patching holes or restoring wall surfaces, if necessary
- PC33.** Educate the client on how to operate the system.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** social environment in the workplace.
- KU2.** good understanding of different types of water purifiers, such as reverse osmosis (RO) systems, activated carbon filters, ultraviolet (UV) disinfection units, and ion exchange systems.
- KU3.** Working of different types of water purifiers



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- KU4.** Determine the most suitable type of purifier based on the water source and the customer's requirements
- KU5.** Telephone etiquette and communication with visitors
- KU6.** The system, processes, timetable and performance expectations of the employer
- KU7.** Personal and workplace hygiene
- KU8.** Good understanding of different types of water purifiers, such as reverse osmosis (RO) systems, activated carbon filters, ultraviolet (UV) disinfection units, and ion exchange systems.
- KU9.** Working of different types of water purifiers
- KU10.** determine the most suitable type of purifier based on the water source and the customer's requirements
- KU11.** Knowledge of plumbing and piping systems
- KU12.** different types of pipes, fittings, valves, and connectors required for installation
- KU13.** Understand electrical safety procedures and have the skills to handle electrical wiring, switches, and outlets
- KU14.** understand the power requirements of the purifier and ensure the electrical connections are properly grounded and protected.
- KU15.** Use of necessary tools and equipment for water purifier installation
- KU16.** safety protocols, including wearing appropriate personal protective equipment (PPE), following electrical safety guidelines, and ensuring proper disposal of any waste materials generated during installation
- KU17.** have knowledge of troubleshooting common issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate (read, write and speak) in the language being used to in the workplace
- GS2.** discuss task lists, schedules, and work-load of the household with co- workers (if any)
- GS3.** use knowledge of numeracy
- GS4.** make decisions pertaining to the concerned area of work
- GS5.** manage relationships with employers
- GS6.** follow basic work ethics such as punctuality, discipline, and regularity
- GS7.** take decisions pertaining to the concerned area of work
- GS8.** utilize existing resources economically
- GS9.** plan out the given/ available tasks
- GS10.** respond to emergency situations/ incidents as per process
- GS11.** use inclusive, gender sensitive language
- GS12.** adjust communication styles to reflect gender sensitivity
- GS13.** make decisions pertaining to the concerned area of work



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Pre-installation activities</i>	13	45	-	-
PC1. Evaluate the quality of the water source intended for purification	1	3	-	-
PC2. select the most suitable water purifier system for specific needs	1	4	-	-
PC3. Select the appropriate purification technology and system capacity	1	3	-	-
PC4. Consider factors such as the type of contaminants to be removed, the desired level of purification, water consumption rate, and available space for installation. Properly sizing the system ensures optimal performance and efficiency.	1	3	-	-
PC5. Evaluate the available options and select the most suitable water purifier for your needs..	-	2	-	-
PC6. Determine if any permits or approvals are required for installing a water purifier system in your area	1	3	-	-
PC7. Follow regulations or codes that need to be followed	1	3	-	-
PC8. Contact the appropriate local authorities or agencies to understand the legal requirements and ensure compliance if any	1	3	-	-
PC9. Identify the best location for installing the water purifier system.	-	3	-	-
PC10. Explain factors such as proximity to the water source, access to plumbing connections, availability of electrical outlets (if required), drainage options, and convenient access for maintenance and filter replacements	1	3	-	-
PC11. Ensure that there is adequate space around the installation location for easy access during installation and future maintenance activities	1	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Allow for sufficient clearance to connect plumbing lines, electrical connections, and access the system for servicing and heat dissipation.	1	3	-	-
PC13. Ensure that these utilities are functioning properly and meet the requirements of the water purifier system being installed.	1	3	-	-
PC14. Understand the installation instructions and guidelines provided by the manufacturer of the water purifier system. These instructions may include specific requirements, recommendations, or precautions that need to be followed during installation	1	3	-	-
PC15. Ensure that all required components and materials for installation, such as plumbing fittings, connectors, pipes, electrical wiring, and mounting hardware, are procured and readily available before starting the installation process.	1	3	-	-
<i>Site Preparation</i>	2	20	-	-
PC16. Prepare the installation site by ensuring it is clean, clear of any obstructions, and suitable for accommodating the water purifier system.	-	5	-	-
PC17. Remove debris, organize the area, and provide sufficient space for the system and associated components	1	3	-	-
PC18. Locate and shut off the main water supply	1	3	-	-
PC19. Connect the water purifier system to the existing plumbing lines.	-	5	-	-
PC20. Mount the water purifier unit securely	-	4	-	-
<i>Electrical Connections</i>	4	6	-	-
PC21. Ensure proper wiring is in place.	2	3	-	-
PC22. Connect power cables, and install switches or outlets as per the manufacturer's guidelines.	2	3	-	-
<i>Installing Pre-Filters and Post-Filters</i>	16	44	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Install any pre-filters or post-filters that are part of the water purifier system.	1	4	-	-
PC24. Follow the manufacturer's instructions for their installation.	1	4	-	-
PC25. Set up the necessary drainage system for the water purifier unit, especially if it utilizes technologies such as reverse osmosis (RO).	1	4	-	-
PC26. Conduct testing the system to ensure its proper functionality.	1	4	-	-
PC27. Perform calibration of any necessary settings, such as flow rate or pressure regulators, is performed at this stage.	1	4	-	-
PC28. Provide the user with instructions on operating and maintaining the water purifier system.	1	4	-	-
PC29. Explain how to use the control panel (if applicable), replace filters, perform routine maintenance, and troubleshoot common issues.	2	4	-	-
PC30. Demonstrate proper usage to ensure the user's understanding.	2	4	-	-
PC31. After the installation is finalized, clean up the installation area, removing any debris, packaging materials, or tools used during the process.	2	4	-	-
PC32. Restore any disrupted areas, such as patching holes or restoring wall surfaces, if necessary	2	4	-	-
PC33. Educate the client on how to operate the system.	2	4	-	-
NOS Total	35	115	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	NIO/PSC/N0101
NOS Name	Install water purifier system (Domestic and Commercial)
Sector	Plumbing
Sub-Sector	Engineering and Technology
Occupation	Water purifier technician
NSQF Level	3.5
Credits	4
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2028
NSQC Clearance Date	23/06/2023



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NIO/PSC/N0102: Perform service and maintenance of Water purifier system (Domestic and Commercial)

Description

This OS unit talks about maintenance and troubleshooting of water purifier system

Scope

The scope covers the following :

- This unit/task covers the following: Routine Maintenance of Water purifier system and fault finding and Troubleshooting of Water purifier system

Elements and Performance Criteria

Routine Maintenance of Water purifier system and fault finding

To be competent, the user/individual on the job must be able to:

- PC1.** Monitor the system's pressure to ensure it is within the recommended range.
- PC2.** check the electrical connections, cables, and plugs for any signs of wear, damage, or loose connections.
- PC3.** Identify and check codes or alarms, and address and resolve any identified issues accordingly
- PC4.** Maintain proper documentation of the maintenance activities
- PC5.** Obtain feedback from the users regarding the water quality, system reliability, and overall performance should be considered to gauge the effectiveness of the maintenance process.
- PC6.** Utilize diagnostic tools, if available, to assess the system's performance and identify specific faults or malfunctions.
- PC7.** Follow appropriate steps to repair or replace the faulty component or system part.
- PC8.** Follow the manufacturer's guidelines or consult a professional if necessary.

Troubleshooting of Water purifier system

To be competent, the user/individual on the job must be able to:

- PC9.** Carry out thorough cleaning of water purifier to remove any dirt, sediment, or debris that may accumulate over time.
- PC10.** Evaluate water quality by testing the water for parameters such as pH, total dissolved solids (TDS), microbial contamination, and chemical impurities.
- PC11.** Ensure that the flow rate of the purified water should be within the acceptable range specified by the manufacturer.
- PC12.** Check the water purifier for any leaks or drips during maintenance.
- PC13.** identify faults and repair promptly to prevent water wastage and system damage.
- PC14.** conduct tests to ensure that the system is functioning properly and meets the required performance criteria mentioned earlier.
- PC15.** conduct functional tests on the system to identify any abnormal behavior or malfunctions.



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Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** social environment in the workplace.
- KU2.** telephone etiquettes and communication with visitors
- KU3.** the system, processes, timetable and performance expectations of the employer
- KU4.** personal and workplace hygiene
- KU5.** good understanding of different types of water purifiers, such as reverse osmosis (RO) systems, activated carbon filters, ultraviolet (UV) disinfection units, and ion exchange systems.
- KU6.** Working of different types of water purifiers
- KU7.** determine the most suitable type
- KU8.** Knowledge of plumbing and piping systems
- KU9.** different types of pipes, fittings, valves, and connectors required for installation
- KU10.** Understand electrical safety procedures and have the skills to handle electrical wiring, switches, and outlets
- KU11.** understand the power requirements of the purifier and ensure the electrical connections are properly grounded and protected.
- KU12.** Use of necessary tools and equipment for water purifier installation
- KU13.**
 - safety protocols, including wearing appropriate personal protective equipment (PPE), following electrical safety guidelines, and ensuring proper disposal of any waste materials generated during installation
 - have knowledge of troubleshooting common issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate (read, write and speak) in the language being used to in the workplace
- GS2.** meet and greet others appropriately
- GS3.** build rapport with various stakeholders
- GS4.** take a problem-solving approach to work
- GS5.** treat people with dignity
- GS6.** remain unbiased at work
- GS7.** demonstrate sound work ethics
- GS8.** manage stress at work and work under pressure
- GS9.** discuss task lists, schedules, and work-load
- GS10.** use knowledge of numeracy
- GS11.** make decisions pertaining to the concerned area of work
- GS12.** perform assigned tasks efficiently based on the priority set by the employer
- GS13.** manage relationships with employers
- GS14.** follow basic work ethics such as punctuality, discipline, and regularity



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- GS15.** take decisions pertaining to the concerned area of work
- GS16.** utilise existing resources economically
- GS17.** organise the items in all parts of the house.
- GS18.** plan out the given/ available tasks
- GS19.** respond to emergency situations/ incidents as per process
- GS20.** use inclusive, gender sensitive language
- GS21.** adjust communication styles to reflect gender sensitivity



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Routine Maintenance of Water purifier system and fault finding</i>	8	24	-	-
PC1. Monitor the system's pressure to ensure it is within the recommended range.	1	3	-	-
PC2. check the electrical connections, cables, and plugs for any signs of wear, damage, or loose connections.	1	3	-	-
PC3. Identify and check codes or alarms, and address and resolve any identified issues accordingly	1	3	-	-
PC4. Maintain proper documentation of the maintenance activities	1	3	-	-
PC5. Obtain feedback from the users regarding the water quality, system reliability, and overall performance should be considered to gauge the effectiveness of the maintenance process.	1	3	-	-
PC6. Utilize diagnostic tools, if available, to assess the system's performance and identify specific faults or malfunctions.	1	3	-	-
PC7. Follow appropriate steps to repair or replace the faulty component or system part.	1	3	-	-
PC8. Follow the manufacturer's guidelines or consult a professional if necessary.	1	3	-	-
<i>Troubleshooting of Water purifier system</i>	7	21	-	-
PC9. Carry out thorough cleaning of water purifier to remove any dirt, sediment, or debris that may accumulate over time.	1	3	-	-
PC10. Evaluate water quality by testing the water for parameters such as pH, total dissolved solids (TDS), microbial contamination, and chemical impurities.	1	3	-	-
PC11. Ensure that the flow rate of the purified water should be within the acceptable range specified by the manufacturer.	1	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Check the water purifier for any leaks or drips during maintenance.	1	3	-	-
PC13. identify faults and repair promptly to prevent water wastage and system damage.	1	3	-	-
PC14. conduct tests to ensure that the system is functioning properly and meets the required performance criteria mentioned earlier.	1	3	-	-
PC15. conduct functional tests on the system to identify any abnormal behavior or malfunctions.	1	3	-	-
NOS Total	15	45	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	NIO/PSC/N0102
NOS Name	Perform service and maintenance of Water purifier system (Domestic and Commercial)
Sector	Plumbing
Sub-Sector	Engineering and Technology
Occupation	Water purifier technician
NSQF Level	3.5
Credits	3
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2028
NSQC Clearance Date	23/06/2023



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NIO/PSC/N0103: Effective Resource Utilization

Description

This OS unit is about the knowledge and skills required to utilize resources effectively.

Scope

The scope covers the following :

- The scope covers the following:
- Planning and adopting a systematic approach and Use of Technology for waste management.

Elements and Performance Criteria

Planning and adopting a systematic approach

To be competent, the user/individual on the job must be able to:

- PC1.** Determine the importance and need of effective utilization of resources in the water purifier system.
- PC2.** Suggest customization of the water purifier unit according to the quality of water at work place.
- PC3.** Identifying the quality of raw water available and using the necessary pre-filter stage in the purifier system.

Use of Technology for waste management

To be competent, the user/individual on the job must be able to:

- PC4.** Choosing energy efficient electrical parts like booster pump, power supply, UV light
- PC5.** Explaining different utilization of discarded water from outlet of water purifier system.
- PC6.** Plan and apply addition provisions with purifier system for using discarded water in best possible manner

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** social environment in the workplace.
- KU2.** telephone etiquettes and communication with visitors
- KU3.** the system, processes, timetable and performance expectations of the employer
- KU4.** personal and workplace hygiene
- KU5.** good understanding of different types of water purifiers, such as reverse osmosis (RO) systems, activated carbon filters, ultraviolet (UV) disinfection units, and ion exchange systems.
- KU6.** Working of different types of water purifiers
- KU7.** determine the most suitable type of purifier based on the water source and the customer's requirements.
- KU8.** Knowledge of plumbing and piping systems



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- KU9.** different types of pipes, fittings, valves, and connectors required for installation.
- KU10.** Understand electrical safety procedures and have the skills to handle electrical wiring, switches, and outlets
- KU11.** understand the power requirements of the purifier and ensure the electrical connections are properly grounded and protected.
- KU12.** Use of necessary tools and equipment for water purifier installation
- KU13.** safety protocols, including wearing appropriate personal protective equipment (PPE), following electrical safety guidelines, and ensuring
- KU14.** proper disposal of any waste materials generated during installation
- KU15.** have knowledge of troubleshooting common issues

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate (read, write and speak) in the language being used to in the workplace
- GS2.** discuss task lists, schedules, and work-load of the household with co- workers (if any)
- GS3.** use knowledge of numeracy
- GS4.** make decisions pertaining to the concerned area of work
- GS5.** perform assigned tasks efficiently based on priority set by the employer
- GS6.** manage relationships with employers
- GS7.** follow basic work ethics such as punctuality, discipline, and regularity
- GS8.** take decisions pertaining to the concerned area of work
- GS9.** utilise existing resources economically
- GS10.** plan out the given/ available tasks
- GS11.** respond to emergency situations/ incidents as per process
- GS12.** use inclusive, gender sensitive language
- GS13.** adjust communication styles to reflect gender sensitivity



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Planning and adopting a systematic approach</i>	10	15	-	-
PC1. Determine the importance and need of effective utilization of resources in the water purifier system.	4	-	-	-
PC2. Suggest customization of the water purifier unit according to the quality of water at work place.	4	7	-	-
PC3. Identifying the quality of raw water available and using the necessary pre-filter stage in the purifier system.	2	8	-	-
<i>Use of Technology for waste management</i>	5	20	-	-
PC4. Choosing energy efficient electrical parts like booster pump, power supply, UV light	2	7	-	-
PC5. Explaining different utilization of discarded water from outlet of water purifier system.	2	7	-	-
PC6. Plan and apply addition provisions with purifier system for using discarded water in best possible manner	1	6	-	-
NOS Total	15	35	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	NIO/PSC/N0103
NOS Name	Effective Resource Utilization
Sector	Plumbing
Sub-Sector	Engineering and Technology
Occupation	Water purifier technician
NSQF Level	3.5
Credits	2
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2028
NSQC Clearance Date	23/06/2023



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NIO/PSC/N0104: Maintain health and safety at work place

Description

This OS unit is about the maintaining health and safety at workplace.

Scope

The scope covers the following :

- The scope covers the following:
- safety of self, co-worker/water purifier unit/ workplace and safe disposal of used part

Elements and Performance Criteria

Safety of self, co-worker/water purifier unit/ workplace and safe disposal of used part

To be competent, the user/individual on the job must be able to:

- PC1.** identify any hazards and deal with them in safe and competent manner within the limits of one's authority
- PC2.** identify and report breaches in health, safety, and security to the designated person
- PC3.** label and store used parts in designated containers until they can be properly disposed of.
- PC4.** respond to different emergency situations in terms of immediate action and reporting
- PC5.** follow environment friendly practices at home/ workplace to minimise pollution of air, water and earth.
- PC6.** use appropriate cleaning gear/ PPE for waste disposal as required
- PC7.** clear, collect and segregate waste according to their types
- PC8.** reduce the volume of waste through appropriate techniques.
- PC9.** throw waste in appropriate waste container/ assigned bins
- PC10.** change disposable garbage bags when full, and clean the waste bins regularly
- PC11.** keep the house/ work area clean off germs/ insects/ bugs by using appropriate environment-friendly products/ cleaning agents.
- PC12.** clean the place of dust or any particulate matters
- PC13.** ensure ventilation in the household
- PC14.** minimise pollution to all environmental aspects such as, water, earth, air and noise.
- PC15.** conserve materials, natural resources and energy at home/ work area.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** specific requirements with respect to the workplace/ house in terms of cleanliness, safety etc.
- KU2.** unsafe conditions and unsafe acts in households and responses and precautions thereof
- KU3.** different types of breaches in health, safety and security and how and when to report these



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- KU4.** use of cleaning gear fire prevention/suppression, personal protective equipment (PPE)
- KU5.** evacuation procedures with special care for PwD and elders
- KU6.** How to reduce the emission of harmful gases
- KU7.** importance of housekeeping in keeping the workplace safe and secure
- KU8.** ways to contact medical assistance and the emergency services, where ever necessary
- KU9.** different emergency situations and responses thereof
- KU10.** health, safety and accident reporting procedures
- KU11.** government agencies in the areas of safety, health and security and their norms and services
- KU12.** essential services such as water, electricity and their optimum utilization/ conservation/ preservation
- KU13.** different categories of waste and how they should be dealt with
- KU14.** methods/ process of reduction, reuse and recycling different types of waste
- KU15.** ways to keep the areas green and lush through plantations, gardens and plant/ flower pots
- KU16.** health, safety and security practices that help to prevent and control infection
- KU17.** types of pollution, such as air, water, soil and noise etc and their control and reduction
- KU18.** ways to minimize dust, avoid spillages and address spillage
- KU19.** usage of cleaning agents/ chemicals with precaution

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate (read, write and speak) in the language being used to in the workplace
- GS2.** discuss task lists, schedules, and work-load of the household with co- workers (if any)
- GS3.** use knowledge of numeracy
- GS4.** make decisions pertaining to the concerned area of work
- GS5.** perform assigned tasks efficiently based on priority set by the employer
- GS6.** manage relationships with employers
- GS7.** follow basic work ethics such as punctuality, discipline, and regularity
- GS8.** take decisions pertaining to the concerned area of work
- GS9.** utilise existing resources economically
- GS10.** plan out the given/ available tasks
- GS11.** respond to emergency situations/ incidents as per process
- GS12.** use inclusive, gender sensitive language
- GS13.** adjust communication styles to reflect gender sensitivity



Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Safety of self, co-worker/water purifier unit/ workplace and safe disposal of used part</i>	15	35	-	-
PC1. identify any hazards and deal with them in safe and competent manner within the limits of one's authority	1	-	-	-
PC2. identify and report breaches in health, safety, and security to the designated person	1	-	-	-
PC3. label and store used parts in designated containers until they can be properly disposed of.	1	3	-	-
PC4. respond to different emergency situations in terms of immediate action and reporting	1	3	-	-
PC5. follow environment friendly practices at home/ workplace to minimise pollution of air, water and earth.	1	3	-	-
PC6. use appropriate cleaning gear/ PPE for waste disposal as required	1	3	-	-
PC7. clear, collect and segregate waste according to their types	1	3	-	-
PC8. reduce the volume of waste through appropriate techniques.	1	3	-	-
PC9. throw waste in appropriate waste container/ assigned bins	1	2	-	-
PC10. change disposable garbage bags when full, and clean the waste bins regularly	1	3	-	-
PC11. keep the house/ work area clean off germs/ insects/ bugs by using appropriate environment-friendly products/ cleaning agents.	1	3	-	-
PC12. clean the place of dust or any particulate matters	1	3	-	-
PC13. ensure ventilation in the household	1	-	-	-
PC14. minimise pollution to all environmental aspects such as, water, earth, air and noise.	1	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. conserve materials, natural resources and energy at home/ work area.	1	3	-	-
NOS Total	15	35	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	NIO/PSC/N0104
NOS Name	Maintain health and safety at work place
Sector	Plumbing
Sub-Sector	Engineering and Technology
Occupation	Water purifier technician
NSQF Level	3.5
Credits	1
Version	1.0
Last Reviewed Date	23/06/2023
Next Review Date	23/06/2028
NSQC Clearance Date	23/06/2023



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQF Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. The assessment for the theory part will be based on SLM (Self learning material) developed by NIOS for the qualification.
2. Evaluation Department of NIOS will develop question papers for theory part and it will be sent to all the designated examination centers for conduct of examinations.
3. Assessment of practical skill for every learner will be done at training centre/AVIs based on the practical guidelines provided by Vocational Education Department of NIOS.
4. To pass the Qualification, every trainee should score a minimum of 50% in theory and practical.
5. In case of successful passing the theory exam once but not the practical exam, the learners can appear in only practical exam and the marks for the Theory can be carried forward.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to



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successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIO/PSC/N0101.Install water purifier system (Domestic and Commercial)	35	115	0	0	150	35
NIO/PSC/N0102.Perform service and maintenance of Water purifier system (Domestic and Commercial)	15	45	0	0	60	25
NIO/PSC/N0103.Effective Resource Utilization	15	35	0	0	50	15
NIO/PSC/N0104.Maintain health and safety at work place	15	35	0	0	50	15
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	100	260	-	-	360	100



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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.